Sustainable Transportation Coordinator

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***Abstract*—Sustainable mobility in urban university campuses is a key lever to mitigate parking shortages, congestion, and local emissions caused by the predominance of private vehicles. This paper reports on the first stages of the Sustainable Transportation Coordinator project, an ongoing study that analyzes the transportation and parking system of the engineering campuses of Universidad Distrital Francisco José de Caldas and outlines an initial coordination concept for inter-campus carpooling and bicycle sharing in a verified institutional environment. Based on field observations, parking measurements, process documentation and a survey of 51 users, the current phase focuses on systems analysis and high-level architectural guidelines that formalize existing informal carpooling practices, highlight bicycle sharing as the most accepted sustainable alternative, and motivate future design and implementation work in subsequent course workshops.**

***Index Terms*—Carpooling, sustainable mobility, campus transportation, bike-sharing, systems engineering**

I. Introduction

Urban university campuses frequently experience mobility deficits where the predominance of private vehicles saturates available parking, generates queues at access points, and pushes overflow parking onto surrounding public streets. At the Faculty of Engineering of Universidad Distrital Francisco José de Caldas (campuses on Calle 40, Calle 34 and ECCI), these pressures are intensified by limited internal infrastructure, motorcycle-oriented basements, and the systematic occupation of public space by around 90 motorcycles parked daily near the Calle 40 headquarters. This configuration not only constrains the effective capacity of the campuses, but also raises safety concerns for users and neighbours and contributes to a negative environmental impact associated with short urban trips that could be served by more sustainable modes.

Existing studies on campus parking and sustainable mobility in Bogotá report that conventional policies oriented to abundant, low-cost parking tend to produce structural inefficiencies and encourage excessive car use rather than solving access problems. Complementary research on Latin American university campuses shows that coordinated interventions such as carpooling schemes and bicycle promotion can reduce local pollutants and greenhouse gas emissions while improving access for students and staff, and local policy documents such as the Bogotá Sustainable Mobility Plan stress the need for closer alignment between institutional transport management and city-wide sustainability goals.

A detailed systems analysis of the engineering campuses reveals that the transportation and parking system exhibits strong temporal and spatial imbalances: arrivals are heavily concentrated in the early morning (with 55% of users arriving before 8:00 a.m.), internal parking is scarce or absent at some sites, and motorcycle overflow is displaced to surrounding public roads near the Calle 40 headquarters. Field observations, parking measurements and process documentation show that vehicle access is controlled through a combination of a digital registration application and manual decisions by security staff, creating bottlenecks at basement entrances and decoupling recorded data from real-time operational practice.

Survey responses from 51 participants offer a user-level view of how the system currently operates. They show a strong dependence on public transport, targeted use of private vehicles for certain trips, and informal car-sharing arrangements coordinated

through messaging apps, mainly to reduce costs and improve perceived safety. At the same time, respondents report a strong willingness to adopt an institutional bicycle-sharing system and a relevant level of interest in a formalized carpooling alternative, suggesting both behavioural readiness and clear opportunities for intervention.

These empirical findings motivate the Sustainable Transportation Coordinator: a coordination concept that integrates inter-campus carpooling and bicycle sharing within a verified institutional environment. The concept builds on the identified elements, relationships and bottlenecks of the current system and proposes a layered architecture combining a student-facing mobile/web interface, an authentication and reputation core, and back-end services connected to institutional data sources. Subsequent sections describe the analytical methods used, present the main patterns and constraints uncovered in the current mobility system, and outline the key architectural guidelines that emerge from this analysis as a foundation for future development and evaluation.

II. Methods and Materials

A. System Analysis Approach

This study applies a systems analysis approach to the transportation and parking system of the engineering campuses of Universidad Distrital Francisco José de Caldas. The objective is to identify key elements, interactions and bottlenecks that affect mobility, safety and environmental performance, providing an evidence-based foundation for a later coordination concept for shared mobility.

B. Data Collection

Field observations and physical measurements were carried out at the Calle 40 campus, the surrounding Calle 34 area and the ECCI site, characterizing available infrastructure and typical use, including the motorcycle-oriented basement at Calle 40, the presence of around 90 motorcycles parked on public streets near the main entrance, and the limited or null internal parking capacity at the other facilities. In parallel, a structured survey of 51 community members captured user profiles, main modes of transport, arrival times, parking difficulties and willingness to adopt formal carpooling and an institutional bicycle-sharing system, while process documentation described how vehicle access is controlled through a digital registration application and manual decisions by security staff, and how informal carpooling is coordinated through messaging applications.

C. Analytical Focus and Key Findings

The collected information was analysed to identify temporal and spatial patterns, user preferences and opportunities for intervention. Results show strong concentration of arrivals in the early morning (55% of surveyed users before 8:00 a.m.), displacement of parking demand towards the Calle 40 campus and surrounding public roads, and a mobility culture that already combines public transport with informal shared trips but lacks institutional support. At the same time, respondents express high willingness to use an institutional bicycle-sharing system and a relevant level of interest in a formal carpooling alternative, highlighting bicycle sharing and coordinated ride-sharing as promising levers for reducing single-occupancy vehicle use between campuses and motivating the architectural guidelines presented in the next section.

III. Results and Discussion

A. System-level interpretation

The systems analysis reveals a critical misalignment between institutional travel demand and available infrastructure. The finding that 55% of users arrive before 8:00 a.m. generates a synchronized peak load that overwhelms the basement capacity at the Calle 40 campus and is manifested in the observed overflow of approximately 90 motorcycles onto public streets near the main entrance. This pattern confirms that small fluctuations in demand during the morning window can trigger queues at access points and displacement of parking towards surrounding public space.

The proposed layered coordination concept addresses these bottlenecks through two main strategic mechanisms. First, a centralized ride-matching tool for carpooling and inter-campus trips is designed to support better temporal and spatial distribution

of arrivals by aligning shared trips with class schedules and campus destinations, which could smooth the entry flow at the basement and reduce saturation at the Calle 40 headquarters. Second, the transition from unregulated messaging applications to a verified institutional platform directly tackles the trust deficit identified in the diagnostic phase: institutional authentication and reputation modules aim to transform informal shared trips into a traceable and safer service for the university community.

Survey and observational data also indicate a high level of readiness among students for sustainable alternatives. Despite infrastructure constraints, respondents show strong willingness to use an institutional bicycle-sharing system and a relevant interest in formalized carpooling, suggesting that these modes are perceived as effective levers to mitigate the current parking crisis and reduce dependence on private vehicles.

B. Survey Results

Table I summarizes the main findings from the survey applied to 51 members of the engineering community.

TABLE I
SUMMARY OF SURVEY RESULTS (N = 51)

Category	Main result
User profile	98% students
Main transport mode	60.8% public transport
Arrivals before 8:00	55% of respondents
Willingness to bike-sharing	74.5% yes
Interest in formal carpooling	34.3% yes

The survey results reveal that, although most respondents rely on public transport for their daily commute, arrivals are strongly concentrated before 8:00 a.m., which coincides with the periods of greatest saturation in the basement and surrounding streets. The high willingness to use an institutional bicycle-sharing system and the non-negligible interest in formal carpooling indicate behavioural readiness for shared and low-emission modes, but these options are not yet supported by coordinated institutional tools. These patterns directly motivate the focus of the Sustainable Transportation Coordinator on inter-campus bicycle sharing and structured carpooling as complementary strategies to reduce single-occupancy vehicle trips and pressure on limited parking resources.

C. Proposed Coordination Concept

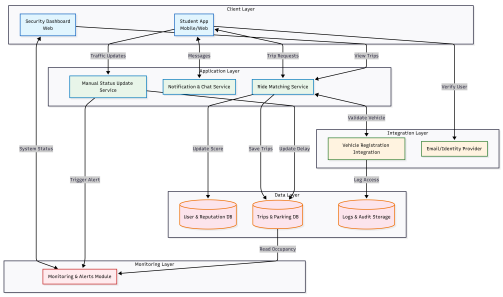

Fig. 1. Layered coordination concept for the Sustainable Transportation Coordinator.

Figure 1 summarizes the layered coordination concept of the Sustainable Transportation Coordinator. The client layer consists of a student mobile/web application and a security dashboard that provide access to ride-matching, notification and status information. The application layer groups core services for ride matching, messaging and manual status updates, while the integration layer connects these services to existing institutional systems for vehicle registration and user authentication. At the data layer, dedicated databases store user and reputation records, trip and parking information, and audit logs, and a monitoring layer consumes these data to generate system status indicators and alerts for operators. Together, these layers reflect the main elements and relationships identified in the system analysis and translate them into a modular architecture for coordinated carpooling and bicycle-sharing management

IV. CONCLUSIONS

The analysis of mobility and parking at the engineering campuses of Universidad Distrital Francisco José de Caldas reveals a pronounced mismatch between user needs and available resources. Concentrated arrival peaks, limited or absent internal parking at some sites and the overflow of motorcycles onto public streets expose structural bottlenecks that

current informal coordination mechanisms, based on messaging applications, are unable to resolve in a safe and systematic way. Survey and observational data confirm both the severity of these constraints and the existence of a socio-technical basis for change, including widespread use of public transport, informal ride-sharing practices and a strong willingness to adopt institutional bicycle-sharing and formal carpooling alternatives.

Within this context, the Sustainable Transportation Coordinator concept provides a structured response grounded in systems analysis. By combining a student-facing platform for bicycle sharing and formalized carpooling with institutional authentication and reputation modules, the proposed layered architecture seeks to reduce single-occupancy vehicle trips, relieve pressure on the Calle 40 campus and improve trust and safety in shared mobility services. The separation into client, application, integration, data and monitoring layers aligns with established systems engineering principles and facilitates future extensions such as additional campuses, new mobility services or integration with city-wide transport data.

The present work constitutes a first step towards a comprehensive smart mobility management solution for the Faculty of Engineering. Future work will focus on specifying and implementing the back-end services in detail, integrating real-time occupancy and registration data, and conducting simulation or pilot studies to quantify potential reductions in parking demand, congestion and emissions under different adoption scenarios. These subsequent phases will make it possible to validate and refine the coordination concept, moving from analytical insight to an operational system that contributes to a more sustainable and efficient urban campus environment.

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